Supplement 4.1 Risk of bias assessment for the methodological quality of the included *in vitro* studies

Reference	Selection bias			Performance bias	Detection bias	Attrition bias	Reporting bias	Confounding bias	Appropriate statistical methods		Appropriate/ controlled exposure (incl. characterization)	Optimal time window used	Statement conflict of interest/fund-ing source	Test substance/treat-ment details	Test organism/system
	Baseline characteristics similarity / appropriate control group selection	Allocation concealment	Randomi-zation	Blinding of researchers	Blinding of outcome assessors	Complete outcome data	Selective outcome data	Account for confounding variables	Sample size determination	Statistical analysis					
Abiko et al. (1998)	+	+	n. a.	n. a.	n. a.	+	+	?	-	-	+	?	?	+	?
Agarwal et al. (2003)	+	+	n. a.	n. a.	n. a.	+	+	?	-	-	+	?	-	+	+
Arima et al. (2019)	?	+	n. a.	n. a.	n. a.	+	+	?	-	-	?	?	?	+	?
Basdra et al. (1995)	?	+	n. a.	n. a.	n. a.	+	+	?	-	-	?	?	-	+	?
Basdra et al. (1996)	?	+	n. a.	n. a.	n. a.	+	+	?	-	-	?	?	-	+	?
Bolcato-Bellemin et al. (2000)	+	+	n. a.	n. a.	n. a.	+	+	?	-	-	?	?	?	+	?
Chang et al. (2015)	?	+	n. a.	n. a.	n. a.	+	+	?	-	-	+	?	?	+	?
Chang et al. (2017)	+	+	n. a.	n. a.	n. a.	+	+	?	-	-	+	?	-	+	+
Chen et al. (2014)	+	+	n. a.	n. a.	n. a.	-	-	?	-	-	+	?	+	?	?
Chen et al. (2015)	+	+	n. a.	n. a.	n. a.	+	+	?	-	-	+	?	+	+	?
Chiba and Mitani (2004)	+	+	n. a.	n. a.	n. a.	+	?	?	-	-	?	?	?	?	?
Cho et al. (2010)	?	+	n. a.	n. a.	n. a.	+	+	?	-	-	?	?	-	+	?
Deschner et al. (2012)	+	+	n. a.	n. a.	n. a.	+	+	?	-	-	+	?	+	+	?
Diercke et al. (2011)	?	+	n. a.	n. a.	n. a.	+	+	?	-	-	+	?	-	+	?
Doi et al. (2003)	?	+	n. a.	n. a.	n. a.	+	+	?	-	-	?	?	?	+	?
Fujihara et al. (2010)	+	+	n. a.	n. a.	n. a.	+	+	?	-	-	?	?	-	+	?
Goto et al. (2011)	?	+	n. a.	n. a.	n. a.	+	?	?	-	-	?	?	+	?	?
Hao et al. (2009)	+	+	n. a.	n. a.	n. a.	+	+	?	-	-	+	?	+	+	+
He et al. (2004)	+	+	n. a.	n. a.	n. a.	+	+	?	-	-	+	?	?	+	?
He et al. (2019)	?	+	n. a.	n. a.	n. a.	+	+	?	-	-	?	+	+	+	+
Howard et al. (1998)	+	+	n. a.	n. a.	n. a.	+	+	?	-	-	+	?	?	+	?
Huelter-Hassler et al. (2017)	+	+	n. a.	n. a.	n. a.	+	+	?	-	-	+	?	+	+	+
Hülter-Hassler et al. (2017)	+	+	n. a.	n. a.	n. a.	+	+	?	-	-	+	?	+	+	?
Jacobs et al. (2013)	?	+	n. a.	n. a.	n. a.	+	+	?	-	-	+	?	+	+	+
Jacobs et al. (2014)	?	+	n. a.	n. a.	n. a.	+	+	?	-	-	+	?	?	+	+
Jacobs et al. (2015)	?	+	n. a.	n. a.	n. a.	+	+	?	-	-	+	?	?	+	+
Jacobs et al. (2018)	?	+	n. a.	n. a.	n. a.	+	?	?	-	-	?	?	+	+	?
Jiang and Hua (2016)	+	+	n. a.	n. a.	n. a.	+	+	?	-	-	?	?	+	+	+
Kaku et al. (2019)	+	+	n. a.	n. a.	n. a.	+	+	?	-	-	+	?	+	+	?
Kanzaki et al. (2006)	?	+	n. a.	n. a.	n. a.	+	?	?	-	-	?	?	?	+	?
Kanzaki et al. (2019)	?	+	n. a.	n. a.	n. a.	+	?	?	-	-	?	?	+	+	?
Kikuiri et al. (2000)	+	+	n. a.	n. a.	n. a.	+	+	?	-	-	+	?	?	+	?
Kim et al. (2007)	?	+	n. a.	n. a.	n. a.	+	?	?	-	-	?	?	-	+	?
Kletsas et al. (2002)	+	+	n. a.	n. a.	n. a.	+	?	?	-	-	+	?	-	+	?
Konstantonis et al. (2014)	?	+	n. a.	n. a.	n. a.	+	?	?	-	-	?	?	+	?	?
Kook and Lee (2012)	+	+	n. a.	n. a.	n. a.	+	+	?	-	-	+	+	?	?	?
Lee et al. (2012)	?	+	n. a.	n. a.	n. a.	+	?	?	-	-	?	?	+	+	?
Lee et al. (2015)	?	+	n. a.	n. a.	n. a.	+	?	?	-	-	?	?	?	+	?
Li et al. (2013)	+	+	n. a.	n. a.	n. a.	+	+	?	-	-	+	?	+	+	?
Li et al. (2014)	+	+	n. a.	n. a.	n. a.	+	+	?	-	-	+	?	+	+	+
Li et al. (2015)	?	+	n. a.	n. a.	n. a.	+	+	?	-	-	?	?	?	+	?

Supplement 4.1 Risk of bias assessment for the methodological quality of the included *in vitro* studies

Reference	Selection bias			Performance bias	Detection bias	Attrition bias	Reporting bias	Confounding bias	Appropriate statistical methods		Appropriate/ controlled exposure (incl. characterization)	Optimal time window used	Statement conflict of interest/fund-ing source	Test substance/treat-ment details	Test organism/system
	Baseline characteristics similarity / appropriate control group selection	Allocation concealment	Randomi-zation	Blinding of researchers	Blinding of outcome assessors	Complete outcome data	Selective outcome data	Account for confounding variables	Sample size determination	Statistical analysis					
Liao and Hua (2013)	+	+	n. a.	n. a.	n. a.	+	+	?	-	-	+	?	+	+	?
Liu et al. (2012)	+	+	n. a.	n. a.	n. a.	+	+	?	-	-	+	?	?	+	?
Liu et al. (2017)	+	+	n. a.	n. a.	n. a.	+	+	?	-	-	+	?	+	+	?
Long et al. (2001)	+	+	n. a.	n. a.	n. a.	+	+	?	-	-	+	?	?	+	+
Long et al. (2002)	+	+	n. a.	n. a.	n. a.	+	+	?	-	-	+	?	?	+	?
Ma et al. (2015)	+	+	n. a.	n. a.	n. a.	+	+	?	-	-	+	?	+	+	+
Matsuda et al. (1998a)	?	+	n. a.	n. a.	n. a.	+	+	?	-	-	?	?	?	+	+
Matsuda et al. (1998b)	?	+	n. a.	n. a.	n. a.	+	+	?	-	-	?	?	?	+	+
Memmert et al. (2019)	?	+	n. a.	n. a.	n. a.	+	+	?	-	-	+	?	+	+	?
Memmert et al. (2020)	+	+	n. a.	n. a.	n. a.	+	+	?	-	-	+	?	+	?	?
Meng et al. (2010)	+	+	n. a.	n. a.	n. a.	+	+	?	-	-	+	?	?	+	+
Miura et al. (2000)	?	+	n. a.	n. a.	n. a.	+	?	?	-	-	?	?	?	?	?
Molina et al. (2001)	?	+	n. a.	n. a.	n. a.	+	+	?	-	-	?	?	?	+	?
Monnouchi et al. (2011)	?	+	n. a.	n. a.	n. a.	+	+	?	-	-	?	?	+	+	?
Monnouchi et al. (2015)	+	+	n. a.	n. a.	n. a.	+	+	?	-	-	+	?	?	+	+
Nakashima et al. (2009)	?	+	n. a.	n. a.	n. a.	+	?	?	-	-	?	?	?	+	+
Narimiya et al. (2017)	?	+	n. a.	n. a.	n. a.	+	?	?	-	-	?	?	?	+	?
Nazet et al. (2020)	?	+	n. a.	n. a.	n. a.	+	+	?	-	+	+	+	+	+	?
Nemoto et al. (2010)	?	+	n. a.	n. a.	n. a.	+	?	?	-	-	?	?	+	+	?
Ngan et al. (1990)	?	+	n. a.	n. a.	n. a.	+	?	?	-	-	?	+	-	+	?
Nogueira et al. (2014a)	?	+	n. a.	n. a.	n. a.	+	?	?	-	-	?	?	+	+	?
Nogueira et al. (2014b)	?	+	n. a.	n. a.	n. a.	+	?	?	-	-	?	?	+	+	?
Nokhbehsaim et al. (2010)	?	+	n. a.	n. a.	n. a.	+	?	?	-	-	?	?	?	?	?
Nokhbehsaim et al. (2011a)	?	+	n. a.	n. a.	n. a.	+	?	?	-	-	?	?	+	+	?
Nokhbehsaim et al. (2011b)	?	+	n. a.	n. a.	n. a.	+	?	?	-	-	?	?	?	+	?
Nokhbehsaim et al. (2012)	?	+	n. a.	n. a.	n. a.	+	+	?	-	-	?	?	+	+	?
Ohzeki et al. (1999)	?	+	n. a.	n. a.	n. a.	+	?	?	-	+	?	?	?	+	?
Ozawa et al. (1997)	?	+	n. a.	n. a.	n. a.	+	?	?	-	-	?	+	?	+	?
Padial-Molina et al. (2013)	+	+	n. a.	n. a.	n. a.	+	+	?	-	-	+	+	+	+	?
Pan et al. (2014)	+	+	n. a.	n. a.	n. a.	+	+	?	-	-	+	?	?	+	?
Papadopoulou et al. (2017)	?	+	n. a.	n. a.	n. a.	+	?	?	-	-	?	?	?	?	?
Papadopoulou et al. (2019)	?	+	n. a.	n. a.	n. a.	+	+	?	-	-	?	?	+	?	?
Pelaez et al. (2017)	+	+	n. a.	n. a.	n. a.	+	+	?	-	+	+	?	?	+	?
Peverali et al. (2001)	+	+	n. a.	n. a.	n. a.	+	+	?	-	-	+	?	-	+	?
Pinkerton et al. (2008)	+	+	n. a.	n. a.	n. a.	+	+	?	-	-	+	?	?	+	?
Qin and Hua (2016)	?	+	n. a.	n. a.	n. a.	?	?	?	-	-	?	?	?	?	?
Rath-Deschner et al. (2009)	+	+	n. a.	n. a.	n. a.	+	+	?	-	-	+	+	+	?	?
Ren et al. (2015)	+	+	n. a.	n. a.	n. a.	+	+	?	-	-	+	?	-	+	?
Ritter et al. (2007)	+	+	n. a.	n. a.	n. a.	+	+	?	-	-	+	?	-	+	?
Saminathan et al. (2012)	?	+	n. a.	n. a.	n. a.	+	?	?	-	-	+	?	?	+	?
Shen et al. (2014)	+	+	n. a.	n. a.	n. a.	+	+	?	-	-	+	?	+	+	?

Supplement 4.1 Risk of bias assessment for the methodological quality of the included *in vitro* studies

Reference	Selection bias			Performance bias	Detection bias	Attrition bias	Reporting bias	Confounding bias	Appropriate statistical methods		Appropriate/ controlled exposure (incl. characterization)	Optimal time window used	Statement conflict of interest/fund-ing source	Test substance/treat-ment details	Test organism/system
	Baseline characteristics similarity / appropriate control group selection	Allocation concealment	Randomi-zation	Blinding of researchers	Blinding of outcome assessors	Complete outcome data	Selective outcome data	Account for confounding variables	Sample size determination	Statistical analysis					
Shimizu et al. (1994)	?	+	n. a.	n. a.	n. a.	+	?	?	-	-	?	?	?	+	?
Shimizu et al. (1995)	?	+	n. a.	n. a.	n. a.	+	?	?	-	-	+	+	?	+	?
Shimizu et al. (1997)	?	+	n. a.	n. a.	n. a.	+	?	?	-	-	?	?	?	+	?
Shimizu et al. (1998)	?	+	n. a.	n. a.	n. a.	+	?	?	-	-	+	?	?	+	?
Spencer and Lallier (2009)	?	+	n. a.	n. a.	n. a.	+	?	?	-	-	?	?	+	+	?
Steinberg et al. (2011)	+	+	n. a.	n. a.	n. a.	+	+	?	-	-	+	?	?	+	?
Sun et al. (2016)	+	+	n. a.	n. a.	n. a.	+	+	?	-	-	+	?	+	+	+
Sun et al. (2017)	+	+	n. a.	n. a.	n. a.	+	+	?	-	-	+	?	?	+	+
Suzuki et al. (2014)	+	+	n. a.	n. a.	n. a.	+	+	?	-	-	+	?	+	+	+
Symmank et al. (2019)	?	+	n. a.	n. a.	n. a.	+	+	?	-	-	?	?	?	+	?
Takano et al. (2009)	+	+	n. a.	n. a.	n. a.	+	?	?	-	-	+	?	?	+	+
Tang et al. (2012)	+	+	n. a.	n. a.	n. a.	+	+	?	-	-	+	?	?	+	?
Tantilertanant et al. (2019a)	+	+	n. a.	n. a.	n. a.	+	+	?	-	-	+	?	+	+	?
Tantilertanant et al. (2019b)	+	+	n. a.	n. a.	n. a.	+	+	?	-	-	+	?	+	+	?
Tsuji et al. (2004)	?	+	n. a.	n. a.	n. a.	+	?	?	-	-	?	?	?	+	?
Tsuruga et al. (2009)	?	+	n. a.	n. a.	n. a.	+	?	?	-	-	?	?	?	+	?
Tsuruga et al. (2012)	?	+	n. a.	n. a.	n. a.	+	?	?	-	-	?	?	+	+	?
Wada et al. (2017)	?	+	n. a.	n. a.	n. a.	+	?	?	-	-	?	?	+	+	?
Wang et al. (2011)	+	+	n. a.	n. a.	n. a.	+	+	?	-	-	+	?	+	+	+
Wang et al. (2013)	+	+	n. a.	n. a.	n. a.	+	+	?	-	-	+	?	?	+	?
Wang et al. (2018)	?	+	n. a.	n. a.	n. a.	+	+	?	-	-	?	?	?	+	?
Wang et al. (2019a)	+	+	n. a.	n. a.	n. a.	+	+	?	-	-	+	?	+	+	?
Wang et al. (2019b)	+	+	n. a.	n. a.	n. a.	+	+	?	-	-	+	?	+	+	?
Wei et al. (2014)	+	+	n. a.	n. a.	n. a.	+	+	?	-	-	+	?	+	+	?
Wei et al. (2015)	+	+	n. a.	n. a.	n. a.	+	+	?	-	+	+	?	+	+	?
Wescott et al. (2007)	+	+	n. a.	n. a.	n. a.	+	+	?	-	-	+	?	?	+	?
Wolf et al. (2014)	+	+	n. a.	n. a.	n. a.	+	?	?	-	-	?	?	?	+	?
Wu et al. (2015)	+	+	n. a.	n. a.	n. a.	+	+	?	-	-	+	?	?	+	?
Wu et al. (2016)	+	+	n. a.	n. a.	n. a.	+	+	?	-	-	+	?	-	+	?
Wu et al. (2017)	+	+	n. a.	n. a.	n. a.	+	+	?	-	-	+	+	+	+	+
Wu et al. (2019a)	+	+	n. a.	n. a.	n. a.	+	+	?	-	-	+	?	+	+	?
Wu et al. (2019b)	+	+	n. a.	n. a.	n. a.	+	+	?	-	-	+	+	+	+	+
Xu et al. (2011)	+	+	n. a.	n. a.	n. a.	+	+	?	-	-	+	?	?	+	+
Xu et al. (2012)	+	+	n. a.	n. a.	n. a.	+	+	?	-	-	+	?	+	+	+
Xu et al. (2015)	?	+	n. a.	n. a.	n. a.	+	?	?	-	-	?	?	?	+	?
Xu et al. (2017)	+	+	n. a.	n. a.	n. a.	+	+	?	-	-	+	?	?	+	+
Yamaguchi and Shimizu (1994)	?	+	n. a.	n. a.	n. a.	+	?	?	-	-	?	?	-	+	?
Yamaguchi et al. (1994)	?	+	n. a.	n. a.	n. a.	+	?	?	-	-	?	+	?	+	?
Yamaguchi et al. (1996)	?	+	n. a.	n. a.	n. a.	+	?	?	-	-	?	?	?	+	?
Yamaguchi et al. (1997)	?	+	n. a.	n. a.	n. a.	+	?	?	-	-	?	?	?	+	?
Yamaguchi et al. (2002)	+	+	n. a.	n. a.	n. a.	+	+	?	-	-	+	?	?	+	?

Supplement 4.1 Risk of bias assessment for the methodological quality of the included *in vitro* studies

Reference	Selection bias			Performance bias	Detection bias	Attrition bias	Reporting bias	Confounding bias	Appropriate statistical methods		Appropriate/ controlled exposure (incl. characterization)	Optimal time window used	Statement conflict of interest/fund-ing source	Test substance/treat-ment details	Test organism/system
	Baseline characteristics similarity / appropriate control group selection	Allocation concealment	Randomi-zation	Blinding of researchers	Blinding of outcome assessors	Complete outcome data	Selective outcome data	Account for confounding variables	Sample size determination	Statistical analysis					
Yamaguchi et al. (2004)	?	+	n. a.	n. a.	n. a.	+	?	?	-	-	?	?	-	+	?
Yamashiro et al. (2007)	+	+	n. a.	n. a.	n. a.	+	-	?	-	-	+	?	?	+	+
Yang et al. (2006)	?	+	n. a.	n. a.	n. a.	+	?	?	-	-	?	?	?	+	?
Yang et al. (2010)	+	+	n. a.	n. a.	n. a.	+	+	?	-	-	+	?	?	+	?
Yang et al. (2015)	+	+	n. a.	n. a.	n. a.	+	+	?	-	-	+	?	+	+	?
Yang et al. (2016)	+	+	n. a.	n. a.	n. a.	+	?	?	-	-	+	?	?	+	?
Yang et al. (2018)	+	+	n. a.	n. a.	n. a.	+	+	?	-	-	+	?	+	+	+
Yoshino et al. (2003)	+	+	n. a.	n. a.	n. a.	?	?	?	-	-	+	?	?	+	?
Yu et al. (2018)	+	+	n. a.	n. a.	n. a.	+	?	?	-	-	+	?	?	+	?
Yuda et al. (2015)	?	+	n. a.	n. a.	n. a.	+	?	?	-	-	?	?	?	+	?
Zhao et al. (2016)	+	+	n. a.	n. a.	n. a.	+	+	?	-	-	+	?	+	+	+
Zhao et al. (2017)	+	+	n. a.	n. a.	n. a.	+	+	?	-	-	+	+	+	?	+
Zhuang et al. (2019)	+	+	n. a.	n. a.	n. a.	+	+	?	-	-	+	+	+	+	?
Ziegler et al. (2010)	+	+	n. a.	n. a.	n. a.	+	+	?	-	-	+	?	?	+	?

Summary	Selection bias			Perfor-mance bias	Detection bias	Attrition bias	Reporting bias	Confounding bias	Appropriate statistical methods		Appropriate/ controlled exposure (incl. characterization)	Optimal time window used	Statement conflict of interest/funding source	Test substance/ treatment details	Test organism/ system
	Baseline characteristics similarity / appropriate control group selection	Allocation concealment	Randomi-zation	Blinding of researchers	Blinding of outcome assessors	Complete outcome data	Selective outcome data	Account for confounding variables	Sample size determination	Statistical analysis					
Low risk of bias ("+")	75 (55%)	137 (100%)	0 (n.a.)	0 (n.a.)	0 (n.a.)	134 (98%)	88 (64%)	0 (0%)	0 (0%)	4 (3%)	80 (58%)	13 (9%)	54 (39%)	126 (92%)	31 (23%)
Number of "?"	62 (45%)	0 (0%)	0 (n.a.)	0 (n.a.)	0 (n.a.)	2 (1%)	47 (34%)	137 (100%)	0 (0%)	0 (0%)	57 (42%)	124 (91%)	67 (49%)	11 (8%)	106 (77%)
Number of "-"	0 (0%)	0 (0%)	0 (n.a.)	0 (n.a.)	0 (n.a.)	1 (1%)	2 (1%)	0 (0%)	137 (100%)	133 (97%)	0 (0%)	00 (0%)	16 (12%)	0 (0%)	0 (0%)
Sum	**137**	**137**	**0**	**0**	**0**	**137**	**137**	**137**	**137**	**137**	**137**	**137**	**137**	**137**	**137**

Supplement 4.1 Risk of bias assessment for the methodological quality of the included *in vitro* studies

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Supplement 4.2 Risk of bias assessment for the reporting quality of the included *in vitro* studies

Reference	Description of scientific background	Description objective	Justification for model	Study design description	Defined experimental outcomes	Ethical statement	Cell maintenance condition	Description of measurement precision and variability	Statistical analysis	Results description
Abiko et al. (1998)	+	+	?	+	+	-	+	+	+	+
Agarwal et al. (2003)	+	+	?	+	+	+	+	-	-	+
Arima et al. (2019)	+	+	?	+	+	+	+	+	+	+
Basdra et al. (1995)	+	+	?	+	+	-	+	-	-	+
Basdra et al. (1996)	+	+	?	+	+	-	+	-	-	+
Bolcato-Bellemin et al. (2000)	+	+	?	+	+	+	+	+	+	+
Chang et al. (2015)	+	+	?	+	+	+	+	+	+	+
Chang et al. (2017)	+	+	?	+	+	+	+	+	+	+
Chen et al. (2014)	+	?	?	+	-	+	+	?	+	+
Chen et al. (2015)	+	+	?	+	+	+	+	+	+	+
Chiba and Mitani (2004)	+	+	?	+	+	+	+	+	+	+
Cho et al. (2010)	+	+	?	?	?	+	+	?	+	+
Deschner et al. (2012)	+	+	?	+	+	+	+	-	-	+
Diercke et al. (2011)	+	+	?	+	?	+	+	+	+	+
Doi et al. (2003)	+	+	?	+	+	+	+	?	+	+
Fujihara et al. (2010)	+	+	?	+	+	-	?	+	+	+
Goto et al. (2011)	+	?	?	?	?	+	?	+	+	?
Hao et al. (2009)	+	+	?	+	+	+	+	+	+	+
He et al. (2004)	+	+	?	+	+	+	+	?	+	+
He et al. (2019)	+	+	?	+	+	+	+	+	+	+
Howard et al. (1998)	+	+	?	+	+	+	?	-	-	+
Huelter-Hassler et al. (2017)	+	+	+	+	+	+	+	+	+	+
Hülter-Hassler et al. (2017)	+	+	+	+	+	+	?	+	+	+
Jacobs et al. (2013)	+	+	?	+	+	n. a.	+	?	+	+
Jacobs et al. (2014)	+	+	?	+	+	n. a.	+	+	+	+
Jacobs et al. (2015)	+	+	?	+	+	n. a.	+	?	+	+
Jacobs et al. (2018)	+	+	?	+	+	n. a.	+	+	+	+
Jiang and Hua (2016)	+	+	?	+	+	+	?	+	+	+
Kaku et al. (2019)	+	+	?	+	+	+	?	+	+	+
Kanzaki et al. (2006)	+	+	?	+	?	+	?	+	+	+
Kanzaki et al. (2019)	+	+	?	+	?	-	?	+	+	+
Kikuiri et al. (2000)	+	+	?	+	+	-	+	+	+	+
Kim et al. (2007)	+	+	?	+	?	+	+	?	-	+
Kletsas et al. (2002)	+	+	?	+	?	-	+	+	+	+

Supplement 4.2 Risk of bias assessment for the reporting quality of the included *in vitro* studies

Reference	Description of scientific background	Description objective	Justification for model	Study design description	Defined experimental outcomes	Ethical statement	Cell maintenance condition	Description of measurement precision and variability	Statistical analysis	Results description
Konstantonis et al. (2014)	+	+	?	?	?	+	+	?	?	+
Kook and Lee (2012)	+	-	?	?	+	+	+	+	+	+
Lee et al. (2012)	+	+	?	+	?	+	?	?	+	+
Lee et al. (2015)	+	-	?	+	?	-	?	+	+	+
Li et al. (2013)	+	-	?	+	+	+	+	+	+	?
Li et al. (2014)	+	+	+	+	+	+	+	+	+	+
Li et al. (2015)	+	+	?	+	+	-	+	+	+	+
Liao and Hua (2013)	+	+	?	+	+	+	+	+	+	+
Liu et al. (2012)	+	+	?	+	+	+	+	+	+	+
Liu et al. (2017)	+	+	?	+	+	+	+	+	+	+
Long et al. (2001)	+	+	?	+	+	+	+	-	-	+
Long et al. (2002)	+	+	?	+	+	+	+	?	+	+
Ma et al. (2015)	+	+	?	+	+	+	+	+	+	+
Matsuda et al. (1998a)	+	+	?	+	+	-	?	-	-	+
Matsuda et al. (1998b)	+	+	?	+	+	+	?	-	-	+
Memmert et al. (2019)	+	+	?	+	+	+	+	+	+	+
Memmert et al. (2020)	+	+	+	?	+	+	+	+	+	+
Meng et al. (2010)	+	+	?	+	+	-	+	-	+	+
Miura et al. (2000)	+	+	?	+	?	-	+	+	+	?
Molina et al. (2001)	+	+	?	+	+	+	?	-	+	+
Monnouchi et al. (2011)	+	+	?	+	+	+	?	+	+	+
Monnouchi et al. (2015)	+	+	?	+	+	+	+	+	+	+
Nakashima et al. (2009)	+	+	?	+	?	+	+	-	-	?
Narimiya et al. (2017)	+	+	?	+	?	+	+	+	+	+
Nazet et al. (2020)	+	?	?	+	+	+	+	+	+	?
Nemoto et al. (2010)	+	+	?	+	?	+	+	+	+	+
Ngan et al. (1990)	+	+	?	+	?	-	?	?	-	+
Nogueira et al. (2014a)	+	+	?	+	?	+	?	+	+	+
Nogueira et al. (2014b)	+	+	?	+	?	+	?	+	+	+
Nokhbehsaim et al. (2010)	+	+	?	+	?	+	?	+	+	+
Nokhbehsaim et al. (2011a)	+	+	?	+	?	+	?	+	+	+
Nokhbehsaim et al. (2011b)	+	+	?	+	?	+	?	+	+	+
Nokhbehsaim et al. (2012)	+	+	?	+	+	+	?	+	+	+
Ohzeki et al. (1999)	+	+	?	?	?	-	+	+	+	+

Supplement 4.2 Risk of bias assessment for the reporting quality of the included *in vitro* studies

Reference	Description of scientific background	Description objective	Justification for model	Study design description	Defined experimental outcomes	Ethical statement	Cell maintenance condition	Description of measurement precision and variability	Statistical analysis	Results description
Ozawa et al. (1997)	+	+	+	+	?	-	?	+	+	+
Padial-Molina et al. (2013)	+	+	+	+	+	-	?	+	+	+
Pan et al. (2014)	+	+	?	+	+	+	+	+	+	+
Papadopoulou et al. (2017)	+	+	?	?	?	+	?	?	+	+
Papadopoulou et al. (2019)	+	+	?	+	?	+	?	+	+	+
Pelaez et al. (2017)	+	+	?	+	-	+	-	+	+	+
Peverali et al. (2001)	+	+	?	+	+	-	+	-	-	+
Pinkerton et al. (2008)	+	+	?	+	+	+	+	?	+	+
Qin and Hua (2016)	+	+	?	?	?	+	?	+	+	+
Rath-Deschner et al. (2009)	+	+	?	+	+	+	+	+	+	+
Ren et al. (2015)	+	+	?	+	+	-	+	+	+	+
Ritter et al. (2007)	+	+	?	+	+	+	+	?	+	+
Saminathan et al. (2012)	+	+	?	+	?	+	?	+	+	+
Shen et al. (2014)	+	+	?	+	+	+	+	+	+	+
Shimizu et al. (1994)	+	+	?	+	?	-	?	+	+	+
Shimizu et al. (1995)	+	?	+	+	?	-	+	+	+	+
Shimizu et al. (1997)	+	+	?	+	?	-	?	+	+	?
Shimizu et al. (1998)	+	+	+	+	?	-	?	+	+	+
Spencer and Lallier (2009)	+	+	?	+	?	+	?	-	-	-
Steinberg et al. (2011)	+	+	+	+	+	+	+	+	+	+
Sun et al. (2016)	+	+	?	+	+	+	+	+	+	+
Sun et al. (2017)	+	+	?	+	+	+	?	+	+	+
Suzuki et al. (2014)	+	+	?	+	+	+	+	+	+	+
Symmank et al. (2019)	+	+	?	+	+	-	?	?	?	+
Takano et al. (2009)	+	+	?	+	?	+	+	+	+	+
Tang et al. (2012)	+	+	?	+	+	+	+	+	+	+
Tantilertanant et al. (2019a)	+	+	?	+	+	+	+	+	+	+
Tantilertanant et al. (2019b)	+	+	?	+	+	+	+	+	+	+
Tsuji et al. (2004)	+	+	?	+	?	+	?	+	+	+
Tsuruga et al. (2009)	+	+	?	+	?	+	+	-	-	+
Tsuruga et al. (2012)	+	+	?	+	?	+	?	-	-	+
Wada et al. (2017)	+	+	?	+	?	+	?	+	+	+
Wang et al. (2011)	+	+	?	+	+	+	+	?	+	+
Wang et al. (2013)	+	+	?	+	+	+	+	+	+	+

Supplement 4.2 Risk of bias assessment for the reporting quality of the included *in vitro* studies

Reference	Description of scientific background	Description objective	Justification for model	Study design description	Defined experimental outcomes	Ethical statement	Cell maintenance condition	Description of measurement precision and variability	Statistical analysis	Results description
Wang et al. (2018)	+	+	?	?	?	-	?	+	+	?
Wang et al. (2019a)	+	+	?	+	+	+	+	+	+	+
Wang et al. (2019b)	+	+	?	+	+	+	+	+	+	+
Wei et al. (2014)	+	+	?	+	+	+	+	+	+	+
Wei et al. (2015)	+	+	?	+	+	+	+	?	+	+
Wescott et al. (2007)	+	+	?	+	+	+	+	-	-	+
Wolf et al. (2014)	+	+	?	+	?	+	?	+	+	+
Wu et al. (2015)	+	+	?	+	+	+	?	+	+	+
Wu et al. (2016)	+	+	?	+	+	+	+	+	+	+
Wu et al. (2017)	+	+	?	+	+	+	+	+	+	+
Wu et al. (2019a)	+	+	?	+	+	+	+	+	+	+
Wu et al. (2019b)	+	+	+	+	+	+	+	+	+	+
Xu et al. (2011)	+	+	?	+	+	+	+	?	+	+
Xu et al. (2012)	+	+	?	+	+	+	+	+	+	+
Xu et al. (2015)	+	+	?	+	?	+	?	+	+	+
Xu et al. (2017)	+	+	?	+	+	+	+	+	+	+
Yamaguchi and Shimizu (1994)	+	+	?	+	?	-	?	+	+	+
Yamaguchi et al. (1994)	+	+	+	+	?	-	+	+	+	+
Yamaguchi et al. (1996)	+	+	?	+	?	-	?	+	+	+
Yamaguchi et al. (1997)	+	+	?	?	?	-	?	+	+	+
Yamaguchi et al. (2002)	+	+	?	+	+	-	?	-	-	+
Yamaguchi et al. (2004)	+	+	?	+	?	+	?	+	+	?
Yamashiro et al. (2007)	+	+	?	?	+	+	?	?	+	+
Yang et al. (2006)	+	+	?	+	?	-	?	+	+	+
Yang et al. (2010)	+	+	?	+	+	-	+	+	+	+
Yang et al. (2015)	+	+	?	+	+	+	?	+	+	+
Yang et al. (2016)	+	+	?	+	?	-	+	+	+	+
Yang et al. (2018)	+	+	?	+	+	+	+	+	+	+
Yoshino et al. (2003)	+	+	?	+	?	-	+	?	+	+
Yu et al. (2018)	+	+	?	?	?	+	+	+	+	+
Yuda et al. (2015)	+	+	?	+	?	+	?	+	+	+
Zhao et al. (2016)	+	+	?	+	+	+	+	+	+	+
Zhao et al. (2017)	+	+	+	+	+	+	+	+	+	+
Zhuang et al. (2019)	+	+	+	+	?	+	+	+	+	+

Supplement 4.2 Risk of bias assessment for the reporting quality of the included *in vitro* studies

Reference	Description of scientific background	Description objective	Justification for model	Study design description	Defined experimental outcomes	Ethical statement	Cell maintenance condition	Description of measurement precision and variability	Statistical analysis	Results description
Ziegler et al. (2010)	+	?	+	+	+	+	+	-	-	?

Summary	Description of scientific background	Description objective	Justification for model	Study design description	Defined experimental outcomes	Ethical statement	Cell maintenance condition	Description of measurement precision and variability	Statistical analysis	Results description
Low risk of bias ("+")	137 (100%)	129 (94%)	14 (10%)	126 (92%)	85 (62%)	100 (73%)	87 (64%)	99 (72%)	117 (85%)	127 (93%)
Unknown/Incomplete ("?")	0 (0%)	5 (4%)	123 (90%)	11 (8%)	50 (36%)	0 (0%)	49 (36%)	20 (15%)	2 (1%)	9 (7%)
High risk of bias ("-")	0 (0%)	3 (2%)	0 (0%)	0 (0%)	2 (1%)	33(24%)	1 (1%)	18 (13%)	18 (13%)	1 (1%)
Not applicable ("n.a.")	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	4 (3%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)
Sum	**137**	**137**	**137**	**137**	**137**	**137**	**137**	**137**	**137**	**137**

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